A low-cost radio reflector for the 21 cm hydrogen line

I am exploring a concept for a low-cost, large-aperture radio reflector operating at the 21 cm neutral hydrogen (HI) line at 1.420 GHz.

The proposed reflector consists of a **10 × 10 m polyethylene matrix**. The actual reflecting surface is made from a **metallized adhesive plastic film**, die-cut with a 10.5 × 10.5 cm pattern. Selected parts of the pattern are removed, leaving a regular array of metallized elements on the polyethylene surface.

The 10.5 cm element spacing corresponds to approximately **λ/2 at the 21 cm wavelength**:

λ ≈ 21 cm
element pitch ≈ 10.5 cm

The right-hand side of the concept shows the **10 m focal length** configuration.

Preliminary cost estimate

For approximately **500 pieces of 450 × 450 mm metallized adhesive film**:

Item	**Estimated cost**
Die-cutting tool	€500–1,500
Metallized adhesive film	€1,000–3,000
Die-cutting	€500–1,500
Roll/sheet handling	€300–800
Reflective film total	**€2,300–6,800**

The polyethylene substrate itself would add approximately **€2,000–5,000**, depending on thickness, material grade and manufacturing method.

For a lightweight steerable support structure capable of carrying a roughly 10 × 10 m reflector and the feed/instrumentation, I would initially estimate approximately **€30,000–80,000** for the mechanical structure, drive system, bearings, actuators, frame and basic control system.

This gives a preliminary total for the reflector and mechanical structure of approximately:

€35,000–90,000

excluding the RF receiver, low-noise amplifier, spectrometer, foundations, site infrastructure and installation.

Comparison with a conventional professional radio telescope

A conventional 10 m-class steerable radio telescope is in a completely different cost category. Published cost studies give a construction-cost scale of roughly **€260,000 for a 10 m single steerable radio dish**, with substantial uncertainty. Another telescope design study estimated approximately **US$500,000 for a mass-produced 10 m reflector at 30 GHz**, excluding non-recurring engineering, electronics, foundation and site infrastructure; the full first-unit cost can be several times higher.

This suggests that the proposed structure could potentially cost **an order of magnitude less** than a conventional professional 10 m-class steerable radio telescope, if the required pointing accuracy and reflector efficiency can be achieved.

There may be an interesting path toward a **large-aperture, low-cost radio telescope for amateur and educational radio astronomy**.